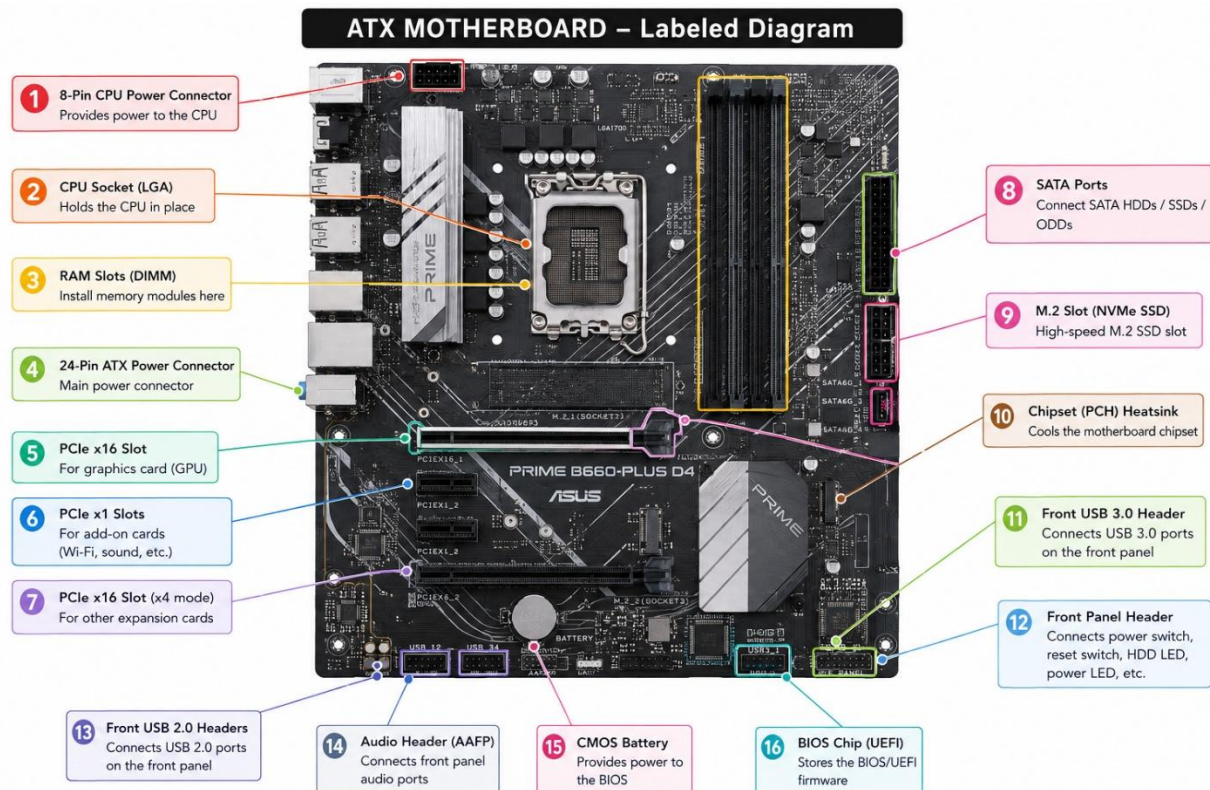


1.

Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



2.

Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

1. LGA1151

- **Company:** Intel
- **Compatible CPU:** Intel Core i5-9400
- **Information:** LGA1151 is an Intel CPU socket. It supports many 8th and 9th Generation Intel processors. It is good for normal work, study, and gaming.

2. AM4

- **Company:** AMD
- **Compatible CPU:** AMD Ryzen 5 5600X
- **Information:** AM4 is an AMD CPU socket. It supports many Ryzen processors. It is popular because it gives good performance and is easy to upgrade.

3. LGA1200

- **Company:** Intel
- **Compatible CPU:** Intel Core i5-10400
- **Information:** LGA1200 is an Intel CPU socket. It supports 10th and 11th Generation Intel processors. It is good for office work, daily use, and gaming.

3.

Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

➤ **CPU (Central Processing Unit)**

The **CPU** is the **brain of the computer**. It reads instructions, processes data, and tells other parts of the computer what to do. It does all the main thinking and calculations.

➤ **Chipset**

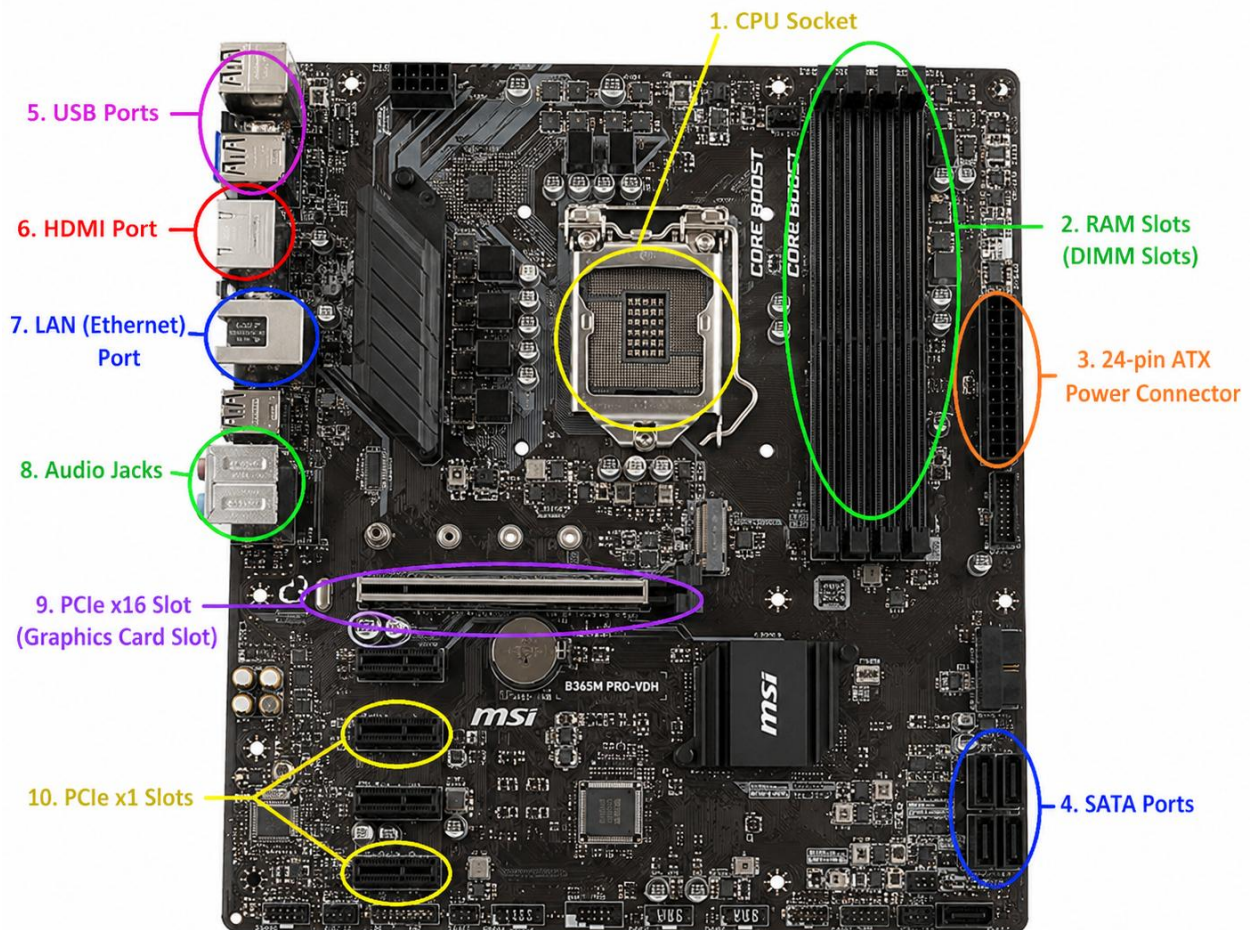
The **chipset** is a part of the motherboard that **helps the CPU communicate** with other parts, such as the RAM, storage (SSD/HDD), USB ports, and graphics card. It makes sure information goes to the correct place.

➤ **How They Work Together**

When you open **Spotify**, the **CPU** starts the app and processes all the instructions. The **chipset** helps the CPU get data from the SSD, load the app into RAM, and send the music and picture data to the screen and speakers. Together, they make Spotify open and play music smoothly.

4.

Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.



➤ **CPU Socket**

Use: This is the place where the CPU (processor) is installed. The CPU does all the main work of the computer.

➤ **RAM Slots**

Use: These slots hold the RAM. RAM stores data that the CPU needs while the computer is running.

➤ **24-pin Power Connector**

Use: This connector gives power from the power supply (SMPS) to the motherboard so all the parts can work.

➤ **SATA Ports**

Use: SATA ports are used to connect storage devices like an HDD or SSD to the motherboard.

➤ **PCIe x16 Slot**

Use: This slot is mainly used to install a graphics card (GPU). It helps the computer run games, videos, and graphics smoothly.